

Zheyuan (Jason) Zhang

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Research Directions

My research focuses on post-training strategies for large language models, with an emphasis on agent harnessing and agentic RL, where the goal is to enable self-evolving, adaptive agent systems that improve their knowledge usage, coordination strategies, and model behaviors over time to dynamic and complex environments robustly and reliably. Specifically, my work spans three areas:

- **Retrieval-augmented Reasoning:** GraphRAG for Reasoning [4, 6, 7, 8]; Long-context Reasoning [11].
- **Agent Harnessing:** Instruction Optimization [5, 9]; Task-aware Routing [3, 4, 9]; Data Collection Workflow [13].
- **Uncertainty-aware Agentic RL:** Uncertainty Quantification [1], Calibration [2]; Intention Clarification [10].

Professional Experience

ByteDance, Inc.

Seattle, WA

Research Scientist - AML AI Foundation (Intern)

Jul. 2026 – Oct. 2026

- Building a next-generation big model as a service platform to serve hundreds of LLMs based applications. Developing and maintaining the big model as a service platform, including offline training/finetuning, online inference, etc.

Amazon, Inc.

Seattle, WA

Applied Scientist - AWS Connect (Intern)

Apr. 2026 – Jul. 2026

- Research on Recoverability-Aware Intervention Learning for Training-Time Rollout Optimization.

Applied Scientist - BRP (Part-time)

Aug. 2025 – Dec. 2025

- Research on uncertainty-aware reinforcement learning and LLM agents for coding and data analytical tasks.

Applied Scientist - BRP (Intern)

May 2025 – Aug. 2025

- Research on LLM agents for code generation. Develop MAPRO [3], a self-evolving prompt optimization framework for multi-agent system. MAPRO was also accepted to AMLC Spotlight and CSS Oral (Amazon Internal Conference).
- Both the coding agents and the MAPRO frameworks are deployed into the *AIDE* (Assistant for Intelligent Deep dive) project, which aims to build an expert system for deep-dive investigations.

IBM Research

Yorktown Heights, NY

Student Researcher (Part-time)

Jun. 2024 – May. 2025

- Research on interpretable and explainable foundation models using graph-based retrieval augmented generation on health-care domains, which leads to two publications to ACL 2025 and KDD 2025. Check this [website](#) for more details of the collaboration project.

Coleridge Initiative, Inc.

New York, NY

Research Scientist

Oct. 2021 – Mar. 2023

- Lead the research project *Food for Thought*: using ML Methods to link food and nutrition databases on a security platform. Responsible for the toolkit and evaluation pipelines and communications between teams. Check [this blog](#) for the competition. Lead the *Unemployment Insurance* project, developing a real-time system to monitor the trend of unemployment rate for IL, IN, TN states.
- Lead the Data Team of Coleridge, responsible for all incoming data ingestion and ETL pipelines.

Associated Research Scientist

Aug. 2020 – Oct. 2021

- Responsible for the *Rich Context* project. This research project aims to identify and extract dataset mentions from full-text social science publications.
- Host the Kaggle Competition *Show US the data*. Responsible for establishing a benchmark for the competition along with technical implementation. A paper is accepted to Harvard Data Science Review.

Junior Research Scientist (Intern)

Oct. 2019 – Aug. 2020

- Work on KG completion and text entity extraction tasks for establishing a large-scale KG for *Rich Context* Competition.

US Ignite, Inc.

New York, NY

Research Fellow (Part-time)

Mar. 2020 – Aug. 2020

- Research on the Smart Monitor for Accelerating Regional Transformation project. Cooperate with senior research fellows to develop a recommendation system to help small business owners find better business locations.

Selected Publications

* Indicate Equal Contribution. To see the full publication list, please visit my google scholar [here](#) .

- [1] **Z. Zhang**, K. Shi, Han. B, Z. Wang, T. Ma, Y. Ye. Why Semantic Entropy Fails: Geometry-Aware and Calibrated Uncertainty for Policy Optimization, 2026, *Arxiv*.
- [2] K. Shi*, **Z. Zhang***, Y. Ye. SAGE: Answer-Conditioned Uncertainty Targets for Verbal Uncertainty Alignment, 2026, *Arxiv*.
- [3] **Z. Zhang***, K. Shi*, Z. Yuan, Z. Wang, T. Ma, K. Murugesan, V. Galassi, C. Zhang, Y. Ye. AgentRouter: A Knowledge-Graph-Guided LLM Router for Collaborative Multi-Agent Question Answering, 2026, *ACL Main*.
- [4] K. Shi*, **Z. Zhang***, Z. Yuan, K. Murugesan, V. Galassi, C. Zhang, Y. Ye. NG-Router: Graph-Supervised Multi-Agent Collaboration for Nutrition Question Answering, 2026, *EACL Main*.
- [5] **Z. Zhang**, L. Ge, H. Li, W. Zhu, C. Zhang, Y. Ye. MAPRO: Recasting Multi-Agent Prompt Optimization as Maximum a Posteriori Inference, 2026, *EACL Findings*.
- [6] **Z. Zhang***, Y. Li*, N. H. L. Le*, Z. Wang, T. Ma, V. Galassi, K. Murugesan, N. Moniz, W. Geyer, N. V. Chawla, C. Zhang, Y. Ye. NGQA: A Nutritional Graph Question Answering Benchmark for Personalized Health-aware Nutritional Reasoning, 2025, *ACL Main*.
- [7] **Z. Zhang**, Z. Wang, T. Ma, V. S. Taneja, S. Nelson, N. H. L. Le, K. Murugesan, M. Ju, N. V. Chawla, C. Zhang, Y. Ye. MOPI-HFRS: A Multi-objective Personalized Health-aware Food Recommendation System with LLM-enhanced Interpretation, 2025, *KDD*.
- [8] **Z. Zhang***, Z. Wang*, S. Hou, E. Hall, L. Bachman, V. Galassi, J. White, N. V. Chawla, C. Zhang, Y. Ye. Diet-ODIN: A Novel Framework for Opioid Misuse Detection with Interpretable Dietary Patterns, 2024, *KDD*.
- [9] J. Huang*, **Z. Zhang***, K. Shi, Y. Ye, C. Zhang. EvolveRouter: Co-Evolving Routing and Prompt for Multi-Agent Question Answering, 2026, *Arxiv*.
- [10] H. Bao, **Z. Zhang**, P. Jing, Z. Yuan, K. Shi, Y. Ye. DRIFT-BENCH : Diagnosing Cooperative Breakdowns in LLM Agents under Input Faults via Multi-Turn Interaction, 2026. *ICML*.
- [11] Y. Li, **Z. Zhang**, T. Ma, Z. Wang, K. Murugesan, C. Zhang, Y. Ye. LongDA: Benchmarking LLM Agents for Long-Document Data Analysis, 2025, *Arxiv*.
- [12] Y. Ye, **Z. Zhang**, et al.. LLMs4All: A Review on Large Language Models for Research and Applications in Academic Disciplines, 2025, *Arxiv*.
- [13] T. Ma, Y. Qian, **Z. Zhang**, Z. Wang, X. Qian, F. Bai, Y. Ding, X. Luo, S. Zhang, K. Murugesan, C. Zhang, Y. Ye. AutoData: A Multi-Agent System for Open Web Data Collection, 2025, *NeurIPS*.
- [14] Z. Wang, **Z. Zhang**, T. Ma, C. Zhang, Y. Ye. Scalable Graph Generative Modeling via Substructure Sequences, 2025, *NeurIPS*.
- [15] Z. Wang, **Z. Zhang**, T. Ma, N. V. Chawla, C. Zhang, Y. Ye. Neural Graph Pattern Machine, 2025, *ICML*.
- [16] Z. Wang, **Z. Zhang**, T. Ma, N. V. Chawla, C. Zhang, Y. Ye. Towards Graph Foundation Models: Learning Generalities Across Graphs via Task-trees, 2025, *ICML*.
- [17] Z. Wang, **Z. Zhang**, C. Zhang, Y. Ye. Computation Tree: A Transferable Pattern Towards Graph Foundation Models, 2024, *NeurIPS*.

Education

University of Notre Dame

Ph.D. in Computer Science

South Bend, IN

Jul. 2023 – Present

- **Research Applications:** Recommendation System, Fraud Detection, Health Intelligence.

New York University, M.S.

Aug. 2019 – Aug. 2020

Wuhan University, B.E.

Sept. 2015 – Jun. 2019

- **Awards:** 1st Class Scholarship 2016 - 2018, Leijun Scholarship 2018

Service

- **Conference Senior Area Chair:** KDD'26
- **Conference Reviewer:** NeurIPS'26, ICML'26, ICLR'26, KDD'26, KDD'25, WWW'25, ReIKDD'24, SDM'25
- **Journal Reviewer:** Transactions on Big Data, Data Mining and Knowledge Discovery, TIST, TMLR, TWEB, PLOS Digital Health.
- **Teaching Assistant:** FA23-CSE-40113, SP24-CSE-40113: Design/Analysis of Algorithms

Invited Talks

- MOPI-HFRS (Sept. 2024) @ IBM Research, Thomas J. Watson Research Center, Yorktown Heights.
- LLM Agents for Code Generation (Jun. 2025) @ Amazon, Seattle.